AI101 MINI-PROJECT · IIT ROPAR

Concrete Strength Prediction using Linear Regression

Team No. 63

Predicting compressive strength (MPa) from the concrete mix

1,030	8	0.63
samples	features	test R^2

REPO LINK - github.com/ajmeerasharath/concrete_strength_prediction_linear_regression_

TEAM & ROLES

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Predicting concrete strength from the mix

WHAT

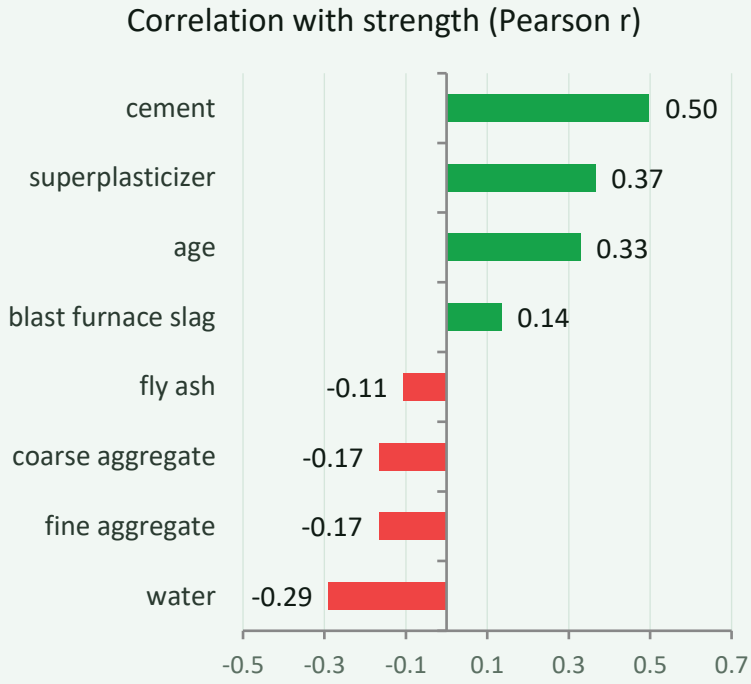
Predict compressive strength (MPa) from 8 inputs: cement, slag, fly ash, water, superplasticizer, coarse and fine aggregate, and age.

WHY

Lab strength tests need weeks of curing. A model gives a quick, cheap estimate before casting.

1,030	9	0
rows	columns	missing

Source: Kaggle, UCI Concrete Compressive Strength



Approach: a four-module pipeline

1

Data

concrete_data.csv

1,030 × 9

2

Analysis

analysis.py

Stats, correlation, heatmap

3

Preprocess

preprocessing.py

Median imputer, 80/20 split

4

Training

training.py

LinearRegression fit on 824 rows

5

Evaluation

evaluation.py

R^2, MAE, RMSE, residuals

MODEL · MULTIPLE LINEAR REGRESSION

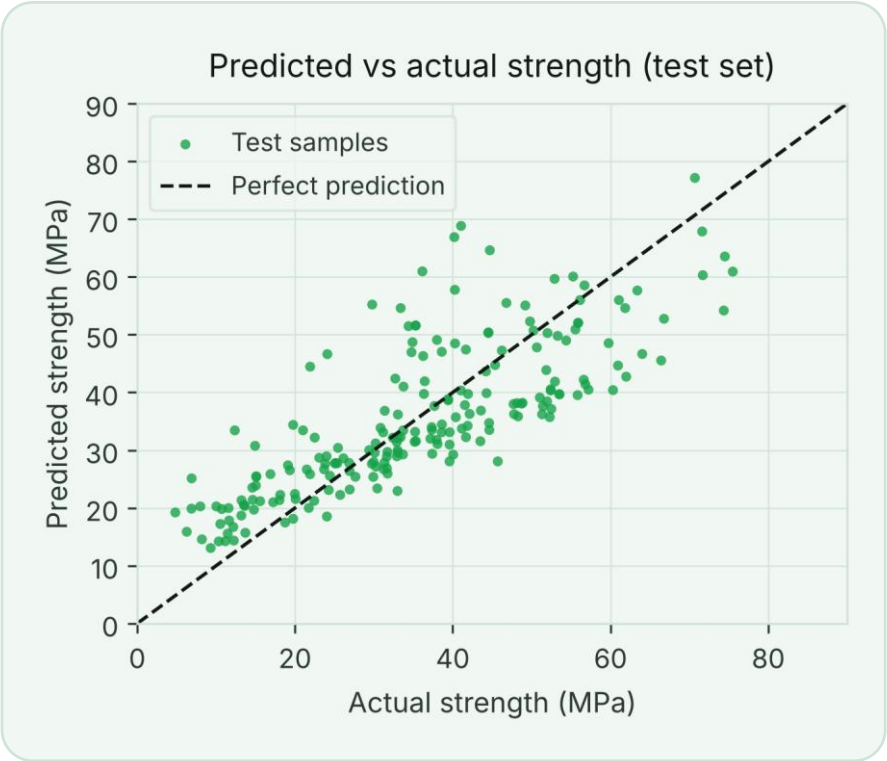
Strength = b0 + b1·x1 + … + b8·x8

Least squares via scikit-learn. Random 80/20 split: 824 train, 206 test, seed 42.

LIBRARIES

- pandas 2.3.3
- NumPy 1.24
- scikit-learn 1.3
- Matplotlib 3.7

Results: R² of 0.63, error about 7.7 MPa



0.63
test R^2

7.75
MAE (MPa)

9.80
RMSE (MPa)

```
$ python Code/main.py
Linear Regression trained!
== MODEL EVALUATION ==
           R2     MAE    RMSE
Training  0.6105 8.3301 10.5188
Testing   0.6276 7.7456  9.7965
```

Train R^2 0.61 vs test R^2 0.63: no sign of overfitting. Predictions rise with actual strength but scatter widely above 60 MPa.

Learnings: wins, limits and roles

What worked

- Modular pipeline: one module per member, runs end to end from main.py
- Train and test R^2 are close, so it generalises
- Coefficients match chemistry: more cement raises strength, more water lowers it

What didn't

- R^2 0.63 leaves 37% of variation unexplained
- Strength vs age is curved; a straight line misses it
- Next: log(age), polynomial terms or Random Forest

Who did what

- Sharath: README, structure and evaluation.py
- Kabir: setup, fresh-clone test and main.py
- Amit: presentation and training.py
- Bhargav: video and analysis.py
- All four wrote and committed code

AI-use disclosure ChatGPT: planning, README, debugging help. Claude: drafted slides and helped with training code. We ran and reviewed all code; no AI generated the dataset or results.